

On the form of Kant's *Critique of Pure Reason*

Eduardo Ramos (Philosophy/UFGA)

1. Introduction

This work aims to present the form of Kant's *Critique of Pure Reason* (CPR) with regard to the exposition of the structural elements that form the plan or general project of this work. To this end, it is necessary to isolate the form from the content of this book, since the latter concerns its results or theses defended, properly speaking, which involve a theory of cognition and a semantic theory.

The general presentation of the *CPR format* will be given here on four main aspects: the internal division of the book, the established objectives, the formulated problems, and the research method used.

2. Internal Division of the *Critique of Pure Reason*

It is commonly known that the internal division of the *Critique of Pure Reason* This serves as the basis for the division of the other two *Critiques* – the *Critique of Practical Reason* and the *Critique of Judgment*. Indeed, the following question naturally arises: where does the internal division of the *Critique of Pure Reason itself originate*? In principle, there are two ways to answer this question: regarding form, one can affirm that the internal division of the *Critique of Pure Reason* imitates the general division of Aristotelian Logic, as conceived by Kant; regarding matter, one can say that Kant imitates Wolff's metaphysical, epistemological, and semantic content embedded in the Transcendental Logic ¹. For the sake of delimitation, only the internal division of this work from a formal point of view will be addressed here².

Regarding the form of Transcendental Logic, Kant maintains, in the *Critique of Pure Reason*, the two main divisions of Aristotelian Logic, as he understood them in his *Jäsche's logic manual*, namely: the Doctrine of Elements and the Doctrine of Method, and, in the first part, the Analytics and the Dialectic. Furthermore, he redistributes in Transcendental Logic the three basic elements contained in these two parts of Aristotle's General Logic, namely: the Doctrine of Concepts, of Judgments, and of Reasoning. In this sense, it can be said that Kant reproduces and adapts the main parts or divisions of General Logic for his Transcendental Logic, including profound revisions to the former, such as the inclusion of an extra-logical topic, which was extracted from Aristotelian Logic (theory of categories) and redirected to form an independent part, called Transcendental Aesthetics, with the

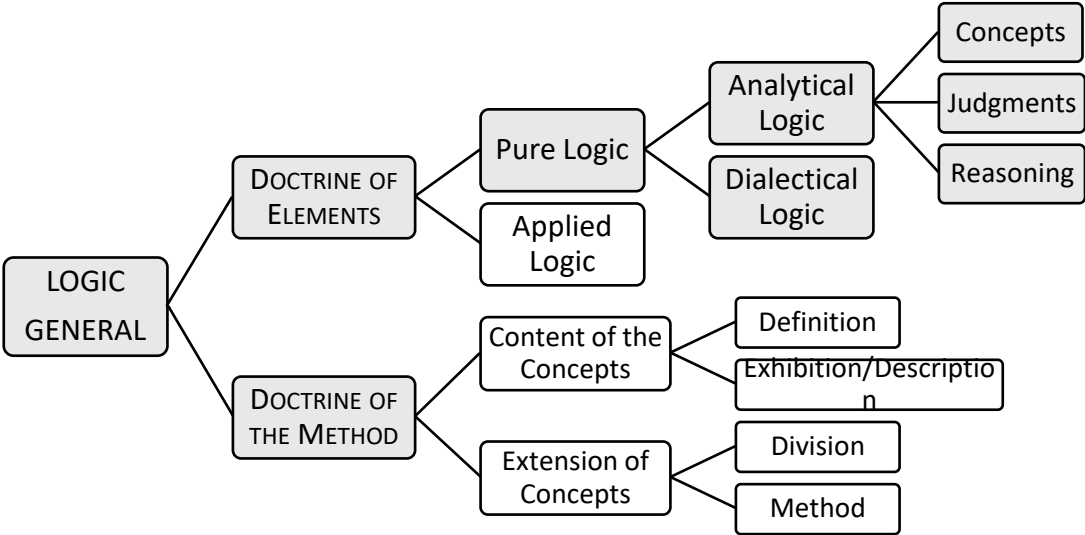
¹ See Bordignon, 2017.

² Longuenesse (2019) also highlights the influence of General Logic on Transcendental Logic regarding the dependence of the table of categories on the table of judgments, but does not present the relationship between these two Logics from the structural perspective that is presented here.

objective of dealing exclusively with space and time – which represents an original contribution of Kant in the *Critique of Pure Reason*.

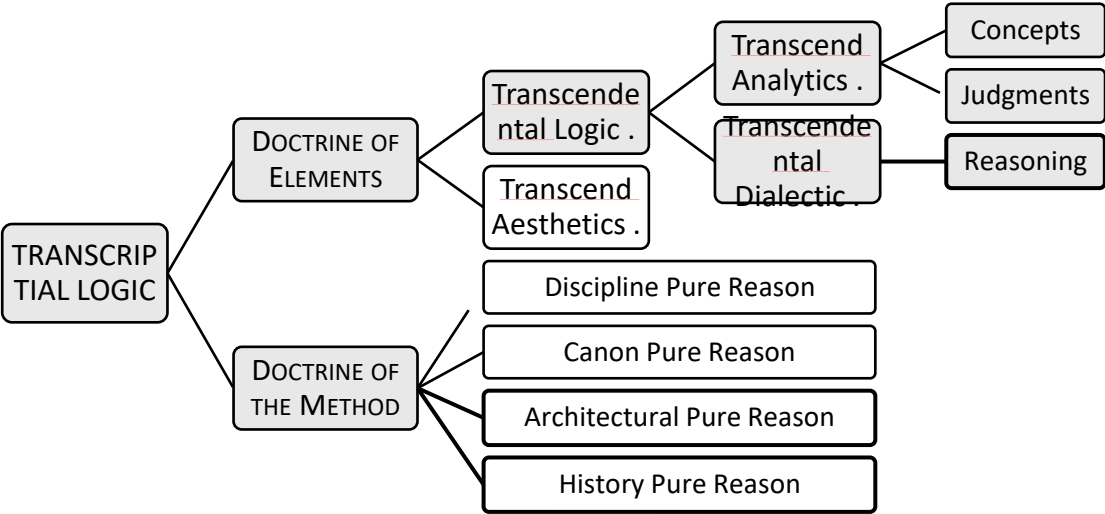
To summarize the exposition, two comparative tables follow, showing the similarity in the division of Aristotle's General Logic and Kant's Transcendental Logic³.

Table 1 shows the division of Aristotle's Logic, as made by Kant in his *Jäsche's logic manual* (1800):



Quadro 1

Figure 2 shows the internal division of Kant's *Critique of Pure Reason*, similar to the division in Aristotelian Logic:



³ See Eduardo Ramos, 2023 and 2025.

Quadro 2

A comparison of these two charts reveals the following results. Firstly, regarding similarity, the parts filled in gray show that the main parts of Aristotelian Logic were imitated or copied by Kant in his Transcendental Logic. In this case, as already pointed out, these are the following divisions: initially, the division into Doctrine of Elements and Doctrine of Method; then, the division of the first into Pure or Transcendental Logic; subsequently, the division of this into Analytic and Dialectic; and finally, the distribution of the Doctrine of Concepts, Judgments, and Reasoning into these last two.

Secondly, regarding the difference, three points can be highlighted. One: as already indicated, in the division of the Doctrine of Elements, Kant replaces Aristotle's Applied Logic with his Transcendental Aesthetic in order to treat the intuition of space and time separately, since in Aristotelian Logic they were considered categories belonging to the Doctrine of Concepts. Two: in Aristotelian Logic, the Analytics contains the doctrine of concepts, judgments, and reasoning, while in Transcendental Logic Kant includes the doctrine of concepts and judgments in his Analytics, and the doctrine of reasoning in his Dialectic. Three: the division of the Doctrine of Method differs in both Logics, which can be understood as an influence of the *Port-Royal Logic* (1682) on Kant, for the following reasons: on the one hand, Aristotle's *Organon* contains many topics scattered throughout its treatises, such as the question of method, definition, hypothesis, thesis, opinion, science, belief, faculties of knowledge, etc.; on the other hand, the *Port-Royal Logic* was the first to separate these topics into a special section of the Logic, called the Doctrine of Method. Following this trend, each logician or philosopher selected the topics they considered relevant to include in their own Doctrine of Method, which Kant imitated in the last part of the *Critique of Pure Reason*, which had the same title⁴.

3. The aims of the *Critique of Pure Reason*

The objectives of a book belong to its form insofar as they generally indicate the author's intention, the intended destination of their work. This is an important formal element for guiding the reader about the overall plan of the work.

Kant establishes two types of objectives to be achieved by the *Critique of Pure Reason*: the general objective and the specific objectives. On the one hand, regarding the general objective, Kant states, in the preface to the 2nd edition, that

⁴ Meier's manual of logic (*Summa of the Doctrine of Reason*, 1752), a book widely used by Kant in his study of Aristotelian Logic, contains in its second part a topic entitled 'On the Method of Learned Science', dedicated to the study of method.

The aim/task [*Geschäft*] of this Critique of Pure Speculative Reason consists in attempting to transform the traditional procedure /method [*Verfahren*] of Metaphysics and thereby promote a complete revolution in it, following the example of geometers and investigators of nature. It is a treatise on method [*Method*] and not a system of science itself (B XII).

In this passage, Kant clearly indicates that the general objective of the *Critique of Pure Reason (CPR)* is to articulate three central questions: method, science, and metaphysics. The correct interpretation of the relationship between these concepts is as follows: Kant's general objective in the first *Critique* is to apply the method of the established sciences to metaphysics in order to transform it into a science ⁵. The reason Kant gives for establishing this general objective is that he considers all metaphysics produced in the past to be an illusion or fantasy due to the lack of an adequate method that would guarantee it the status of a science. Having made this diagnosis, Kant's proposal is simple: identify the method of the established sciences (mathematics and physics), apply it to metaphysics in order to transform it into a science. In short, Kant's general objective in the *CPR* is to construct a scientific metaphysics through the application of the scientific method used by the existing sciences. The execution of the *Critique of Pure Reason* based on the scientific method is the very exemplification of such a scientific metaphysics, which he calls the general metaphysics of nature ⁶.

On the other hand, regarding specific objectives, Kant establishes, in the introduction to the Transcendental Logic (B 82), three goals to be achieved in order to reach his general objective concerning the construction of a scientific metaphysics. The consistent development of these specific objectives is decisive for the success of the general objective itself. The three specific objectives are as follows: to indicate the origin, the objective validity, and the limit of *a priori knowledge*. In general terms, these specific objectives indicate the subjective and objective conditions presupposed in all knowledge, as well as its legitimate and illegitimate use.

Indeed, regarding origin/source, Kant argues that there exists in the mind [*Gemüt*] a class of pure or non-empirical elements that constitute the subjective condition of knowledge, examples of which are intuitions, categories, schemas, "I think," and heuristic principles, which he groups into distinct faculties. Regarding objective validity, he indicates that the legitimate use of these pure elements

⁵ Regarding this issue, Cohen (1918) defends an epistemological interpretation that the goal of the *Critique of Philosophy* is the foundation of science, while Heidegger (1929) defends a foundation of metaphysics. Both are only partially correct because they defend mutually exclusive interpretations (science or metaphysics) when these are in fact inclusive and related.

⁶ Kant clarifies that the *Critique of Pure Reason* is a general metaphysics of nature that deals with the concept of nature in general, whereas his *Metaphysical Principles of Nature* (1786) is a particular mathematics of nature that deals with the concept of matter in general (*PMCN*, A 15-18).

(concepts and principles) is through their reference or application to possible experience or experience in general, without which they are abstract or illusory. Regarding limit/extension/scope, he shows the illegitimate use of these pure elements beyond possible experience, which produces speculative or fanciful conceptions, as does traditional metaphysics.

4. The problems of the *Critique of Pure Reason*

The problem constitutes a formal aspect of the text insofar as it indicates the question to be investigated and answered by the author, and its purpose is to delimit the field of research and to better expose the answer given in the work to judgment.

Kant distinguishes two kinds of problems in the *Critique of Pure Reason* (*CPR*): the general problem and secondary or derived problems. Firstly, the general problem is formulated as follows: “How are synthetic *a priori* judgments possible?” (*CPR*, B 19). In turn, Kant divides the general problem into two parts: “How are they possible” and “synthetic *a priori* judgments” (*Prol.*, A 41). Regarding the second part of the general problem, he understands that in the expression “synthetic *a priori* judgments”, “synthetic” means that it is a judgment that refers to possible or general experience, while “*a priori*” means that such a judgment has a necessary and universal character. In short, the expression “synthetic *a priori* judgment” means a necessary and universal judgment that refers to experience in general. For Kant, synthetic *a priori* judgments are the scientific judgments, properly speaking, which are exemplified in mathematics and physics, as well as in the *Critique of Pure Reason* (defined as a metaphysics of nature) through the pure principles of understanding.

Regarding the first part of the general problem, “how are they possible?”, Kant states that it constitutes its core, since he considers that synthetic *a priori* judgments exist and are given in the consolidated sciences (*Ibidem*). For Kant, the expression “how are they possible?” indicates an investigation that has at least three meanings: 1) the investigation into the conditions of possibility of knowledge (subjective and objective) ; 2) the investigation of the appropriate method for producing synthetic⁷ *a priori* judgments; 3) the unknowability of the relationship between the “synthetic” and the “*a priori*”. The first investigation has a cognitive and semantic nature, the second a methodological character, and the third a skeptical view. Indeed, regarding the first investigation into the general conditions of knowledge, Kant reveals that they are of two kinds: (i) subjective conditions, the development of which presupposes a theory of cognition or of the faculties to expound the class of pure elements (concepts and principles) belonging to the mind of the subject, which are involved in the production of knowledge; (ii) objective

⁷It is worth noting that this investigation of the first part of the general problem connects to two specific objectives of the *CPR*, the origin and the objective validity of knowledge, since both investigate the same object: the subjective and objective conditions of all knowledge.

conditions, the attainment of which involves a semantic theory to deal with the reference of such pure elements to possible or general experience. In turn, regarding the second investigation into the method for producing synthetic *a priori* judgments, Kant reveals that such judgments presuppose an adequate procedure for their production, which ultimately implies the joint satisfaction of the subjective and objective conditions of knowledge. Therefore, for Kant, explaining "how things are possible" implies, on the one hand, an investigation of the subjective and objective conditions of knowledge and, on the other, an investigation of the method, and all these investigations together imply, respectively, the development of a theory of cognition, a semantic theory, and a theory of method⁸. Finally, regarding the investigation into how the relationship between the "synthetic" and the "*a priori*" occurs, Kant confesses an epistemological skepticism in *PMCN* (A 19/29, note), admitting an ignorance in explaining the strange and mysterious isomorphism between the *a priori* laws of nature and contingent phenomena, despite considering that his explanation based on the articulation of the theory of cognition and the semantic theory is better than that of Leibniz (pre-established harmony) and that of Hume (the laws are contingent beliefs).

Secondly, regarding secondary problems, Kant derives four of them from the general problem: 1) how is pure mathematics possible? 2) how is pure physics possible? 3) how is metaphysics possible as a natural disposition? 4) how is metaphysics possible as a science? (*CPR*, B 20-22)⁹. The first two secondary problems deal with science and the last two with metaphysics, with the last problem, concerning the possibility of a scientific metaphysics, being the most relevant of all for Kant, for which the *CPR* is given as an answer and example.

4. The Method of the *Critique of Pure Reason*

The method relates to the form of a work insofar as it reveals the procedure or technique used to investigate the object of study. It is an important element in judging on the consistency and limitations of the approach to the object taken through a particular investigative perspective.

Kant declares that the *Critique of Pure Reason* is a "treatise on method" (B XXII), in which he elaborates a "doctrine of method" (B 108-9) and, in the last part of this book, presents a "Transcendental Doctrine of Method" (B 735-6). Despite this, he does not make clear in these passages which method he uses in constructing this work, although he implicitly uses a particular method. However, in the pre-critical phase¹⁰, Kant explicitly reflects on the question of method and concludes that metaphysics should imitate the analytical and synthetic methods of physics

⁸The exposition of cognitive theory and semantic theory is part of the content of the *CRPu*.

⁹It is worth noting that the division of the *Prolegomena* follows this division of the four secondary problems.

¹⁰ See *Enquiry into the Evidence of the Principles of Natural and Moral Theology* (1764) and *Form and Principles of the Sensible and Intelligible World* (1770).

and mathematics¹¹. For him, the analytical method consists of investigating conditions, principles, or simple parts/elements. The synthetic method, on the other hand, investigates the reference of these principles, that is, the conditioned, the consequence, the composition of the whole.

In the preface to the *Critique of Pure Reason*, when discussing the 'Copernican revolution', Kant indirectly acknowledges the use of the analytic-synthetic method by mathematics and physics and its possible application to metaphysics:

I should think that the examples of Mathematics and Natural Science, which became what they are now through a revolution carried out all at once, would be sufficiently remarkable to make one reflect upon the essential elements of the transformation in the way of thinking [*Denkart*] that was so advantageous to them and, insofar as their analogy with Metaphysics as knowledge of reason allows, to imitate them in this at least as an attempt (B XVI).

In a certain sense, it can be said that, for Kant, the true meaning of the Copernican revolution is fundamentally methodological in nature, as it aims to reveal the method used by mathematics and physics to become consolidated sciences, this being identified with the analytical and synthetic methods. His examples are quite clear regarding the use of this method by these sciences: in mathematics, concepts are conceived by the analytical method and then referred to pure intuition by the synthetic method; in physics, the laws of nature are formulated by the analytical method and subsequently referred to experience by the synthetic method. Therefore, the great challenge that Kant proposes to himself in the *Critique of Pure Reason* is to apply this analytical-synthetic method to metaphysics in order to construct it as a science, that is, to produce a scientific metaphysics.

In general terms, Kant's application of the analytical-synthetic method in the *Critique of Pure Reason* is done as follows. The analytical method is applied in his metaphysical deduction or exposition in order to investigate the subjective conditions of knowledge (intuitions, categories, etc.), while the synthetic method is used in his transcendental deduction or investigation in order to investigate the reference to the objective conditions of knowledge¹².

5. Final considerations

In general terms, it can be concluded that the form of Kant's *Critique of Pure Reason* shows a structural vision of this book through the general exposition of the plan or project conceived in its execution, according to certain technical

¹¹ Cf. Scherer (2019).

¹² A similar interpretation can be found in Cohen (1918) and Loparic (2000).

requirements of a textual nature that involve at least four main elements, which have been presented here in general terms, namely: the division of the work, its general and specific objectives, the general and secondary problems, and finally, the method of investigation.

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